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Tribological performances of polymeric plain bearings made of epoxy matrix with UHMWPE, MoS_2, Kevlar and base oil as the fillers

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Abstract

Thermal, mechanical and tribological performances of epoxy-based composites, filled with UHMWPE, MoS_2 and Kevlar™ or base oil in different proportions, were investigated. Friction coefficient and specific wear rate were examined by a block-on-cylinder tribotester in a dry sliding condition under variable contact pressure and sliding velocities. It was found that the composite with in-situ base oil possesses the lowest coefficient of friction (0.056) and highest wear resistance (5.77×10^{-6} mm^3/Nm) at relatively high PV factor (product of contact pressure and sliding velocity) of 37 MPa-m/s when compared to other composites. Kevlar-filled composite provided low wear rate at very high PV factor of 52.5 MPa-m/s.

Keywords

Polymer composite, plain bearing, epoxy, Kevlar composite

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Introduction

Well-designed polymeric bearings, being self-lubricating, work without any external oil or grease, or with minimal liquid lubricants. They are generally used in industries such as textiles, food, pharmaceuticals, paper, ship propeller shaft etc. where lubricating oils are not desirable. Specialized tribological polymers are also used as ball-bearing cages, gears, tires, dry-bearings, non-stick frying pans, slides, shoe soles, flooring, gyroscope gimbals, bearings in human joint prosthetic implants, medical devices, hybrid bushings in diesel fuel injection pumps,[1,2] as well as in space or the nuclear environment where oils will either solidify or evaporate.[3,4] Lightweight, resistance to wear, low friction and anti-corrosion properties of polymers make them suitable for many applications. Some polymers and their composites can operate in the temperature range between −50 °C to 270 °C and this leads to a broader operating range for wider tribological applications.[5,6] PV_{limit} (pressure × velocity) is an essential factor in the design and selection of polymeric bearing materials. Above PV_{limit} the material fails as friction and wear increase unacceptably mainly as a result of frictional heating and increase in the contact area. Thus, PV_{limit} characterization approach is necessary for the design of plain bearings. Performance parameters important from tribological point of view for bearing application are low wear, low friction and high PV_{limit}. The bearing surface temperature is also very critical factor as it can cause deterioration to the polymer bearing materials due to their low thermal conductivity. Surface temperature is a result of the surrounding heat and frictional heat generated. Polymers, in their pure form, are not very suitable for high stress and high temperature applications. Hence, fillers are invariably added to improve the load carrying capacity and thermal conductivity. If properly selected, they also help improve the tribological performances. For tribology, fibres, particulates, and liquids as fillers have been tried. Researchers[7] have reported that polymer bearing containing 55 wt% PPS (polyphenylene sulfide), 25 wt% PTFE, 10 wt% graphite and 10 wt% lead oxide showed coefficient of friction around 0.18 and specific wear rate of 7.4×10^{-7} mm^3/Nm under a load of 133 N at a shaft speed 1.6 m/s (1610 rev/min) (maximum PV factor of 1.2 MPa-m/s). Bearings made of 60% bronze-filled PTFE was tested under 0.50 MPa pressure and speed 0.11 m/s and found to give coefficient of friction of 0.18 and material loss of 2.9×10^{-3} g.[8] Tribological behaviours of polymer bearings

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made of polymers such as polyethylene, polyamide, POM, Bakelite and PTFE were tested at 20 N of normal load and 0.13 m/s of sliding velocity.[9] It was found that the coefficients of friction of these polymer bearings were in the range of 0.15 to 0.25. Specific wear rates of pure polyethylene were 6.43×10^{-6} mm^3/Nm, of polyamide was 4.3×10^{-6} mm^3/Nm, of POM was 4.12×10^{-6} mm^3/Nm, of PTFE was 7.2×10^{-6} mm^3/Nm, and of Bakelite was 7.47 $\times 10^{-6}$ mm^3/Nm.[9] PAEK (Polyaryletherketone) based bearing filled with MoS_2 and tungsten disulphide was tested for 100–500 N load (nominal pressure of 0.5-2.5 MPa) under a speed of 0.5 m/s. The sliding distance was kept as 3.7 km. The specific wear rate of both the bearings was in the range of $2.3–7.3 \times 10^{-6}$ mm^3/Nm for a PV factor range of 0.25–1 MPa m/s and friction was in the range of 0.09–0.04. In comparison, it was found that PAEK-tungsten disulphide bearing showed better tribology than PAEK-MoS_2 bearing.[10] Numerous polymeric and non-polymeric materials are available now for the sliding layers of plain sliding bearings. Tin based alloys are also used in marine shaft bearings under static load.[11] Victrex material (PEEK composites) bearings were tested (Block on ring) at two different pressures and speeds. At 5 MPa and 1 m/s speed, COF was found to be 0.2 and at 7 MPa and 2 m/s sliding speed, COF was found to be 0.6.[12] Various polyetherketone (PEK) family polymers with nano-particle fillers such as MoS_2, CNT and WS_2 have been suggested for future plastic bearings.[13] In this present paper, we have evaluated the tribological and bulk mechanical performances of the epoxy-based polymer bearings filled with UHMWPE particles, MoS_2 particles and Kevlar (aramid) short fibre or base oil droplets and compared their performances. Kevlar has the advantages of excellent tensile and impact strength, anti-wear, however, it has the limitation of low thermal conductivity. MoS_2 is a well-known solid lubricant having low friction property and it also contributes to dissipate the heat generated at the interface. UHMWPE has a self-lubricating feature with excellent wear resistance when compared with most of the known polymers at moderate contact conditions.[14] Researchers[15] have used UHMWPE with 1 wt% graphene Oxide, 1 wt% nano diamonds and 10 wt% short carbon fibres. They found that the hybrid composites have the lowest coefficient of friction of 0.095 and specific wear rate of 1.2×10^{-6} mm^3/Nm at the contact pressure of 5 MPa and speed of 0.13 m/s. In contrast, it was found that by including silicon carbide and silicon nitride in UHMWPE, wear rate increased to the range of 10^{-5} mm^3/Nm and 10^{-3} mm^3/Nm at 10 N load and 60 N load, respectively.[16,17] In our previous works,[18–21] we have tested epoxy-based composites and found excellent tribological performances (specific wear rate in the range of 10^{-7} mm^3/Nm and low coefficient of friction of 0.08–0.1) at a constant PV of 0.611 MPa-m/s in dry condition. In the current paper, we have further extended the tribological test of epoxy composite bearing up to a PV factor of 52.5 MPa-m/s for evaluating safe industrial application of this composite as a bearing material. The novelty of this composite is that it incorporates both the solid/liquid lubricants (UHMWPE, MoS_2 and base oil) and the strengthening fibre (Kevlar). This is a unique combination for achieving high PV limit of the polymer composite bearing.

Methodology

Materials

A commercial epoxy resin AY 103 (Sigma Aldrich) (density 1.15 g/cm^3) curable by amine-based hardener HY 991 (Sigma Aldrich) (density 0.93 g/cm^3) was considered as the matrix material for this work. UHMWPE powder with particle size of 40–48 µm (Sigma Aldrich) and density of 0.94 g/cm^3, and MoS_2 powder with particle size of ≤2 µm (Sigma Aldrich) and density of 5.06 g/cm^3 were used as the solid lubricants. Kevlar 129 (a trade name) of density 1.44 g/cm^3 was used as a strengthening component for the preparation of polymer composite. Length of the Kevlar short fibres used was approximately 10 mm. Liquid lubricant base oil (SN150) of group II and density 0.875 g/cm^3 was added as the in-situ lubricant to the composite.

Polymer composite preparation

Three types of composites were prepared. For the first composite, epoxy monomer liquid (69.67 vol%), amine-based hardener (8.37 vol%), MoS_2 (0.66 vol%) powder and UHMWPE (21.30 vol%) powder were poured in a glass beaker and mixed using magnetic stirrer for 30 min at 60 °C temperature. This was the base composite, and the sample was named as 'MS'.[15] The second composite, named as K, was prepared by adding 2 vol % 'Kevlar 129' of short length 10 mm to the composition of MS. The third composite, named as B, was prepared by mixing 18 vol% of base oil to the composition of MS.[18] Once the homogeneous mixture of composite for each case was ready and trapped air was removed, the mixture was transferred to a Teflon mould and left at room temperature for initial curing for 20 h. For the 'K' sample preparation, the fibres were first placed in the mould and the polymer mixture was then poured. This was done as the fibre mixed polymer mixture had severe flowability issue due to very high viscosity. Sample cured at room temperature was removed from the Teflon mould for high temperature curing at 140 °C for 5 h in an oven. After oven curing, the sample was kept in a desiccator for 12–15 h at room temperature for stabilization before use. All the samples were prepared in the Teflon mould in the form of circular ring of rectangular cross-section. Dimensions of the circular ring were; outer diameter 40 mm, inner diameter 20 mm and thickness 15 mm. Counterface for the tribotests was EN31 steel cube of dimensions 15 mm × 15 mm × 15 mm and roughness (R_a) 0.2 µm. Figure 1(a) is the photo of the mould in which sample was prepared with Kevlar fibres placed homogenously inside and (b) is the image of Kevlar short fibres before they were placed in the mould. Figure 2 shows the images of the prepared polymer composite bearing rings of base composition (MS), with Kevlar short-fibre (K) and with base oil (B).

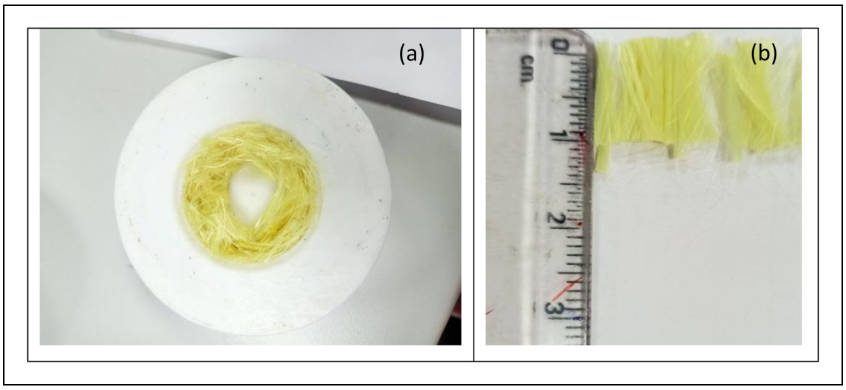

Figure 1. (a) Teflon mould for sample preparation (b) Kevlar fibres before placing them in the mould.

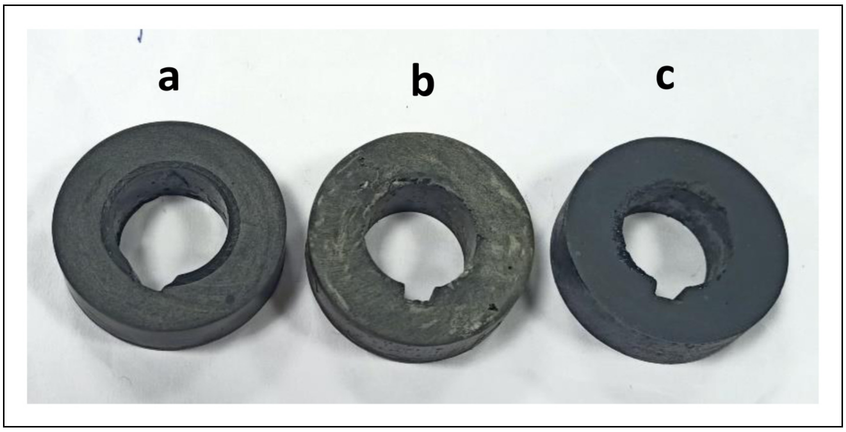

Figure 2. Image of prepared polymer composite (a) MS (b) K (c) B. Colour wise they all had the same black appearance because each contained MoS_2 particles.

Tribology tests

Tribology tests were conducted on a commercial tribometer (TR 31 lubricity tester, EP-LT020120 manufactured by Ducom Ltd India) as shown schematically in Figure 3. This machine (Figure 3) contains a rotating shaft on which the prepared polymer bearing was fixed at one end with a screw and washer. Counterface of steel block was placed on top of the polymer bearing which provided non-conformal contact of block-on-cylinder (line contact). The dead weight was applied as the normal load. Friction force was measured between the circular ring-shaped polymer composite sample and the counterface EN 31 steel block (cube) using a load cell and was recorded with the help of a data acquisition card and personal computer. Normal force capacity of this equipment was from 10 N to 100 N. The applied load on bearing was 20 N, 50 N, 70 N and 90 N, resulting in the maximum Hertzian contact pressures, in MPa, of 9.9, 15.7, 18.5 and 21, respectively. Hertzian contact pressure for line contact was calculated using the below equation applicable for nonconformal line contact.[22]

$$b = \sqrt{\frac{2\ x\ F\left(\left(\frac{1-\nu_1^2}{E_1}\right)+\left(\frac{1-\nu_2^2}{E_2}\right)\right)}{\pi\ x\ l\left(\frac{1}{d_1}+\frac{1}{d_2}\right)}} \qquad (i)$$

$$P_{max} = \frac{2F}{\pi bl} \qquad (ii)$$

where F is the applied normal force in N, ν_1 is the Poisson's ratio of first tribo-pair, ν_2 is the Poisson's ratio of second tribo-pair E_1 and E_2 are elastic modulii of first and second materials of the tribo-pair. d_1 and d_2 are the diameters of curvature of the first and second parts of the tribo-pair. l is the length of cylinder and b is the contact half-width. Since, in the block-on-cylinder geometry, the block is flat surface and hence d_2 is considered as ∞. E_1 and E_2 are used as 4.3 GPa and 215 GPa, respectively, from the previous work.[15]

Shaft rotation rpm was set to achieve the surface sliding speed on the cylinder top surface of 0.5 m/s, 1.0 m/s, 1.5 m/s, 2 m/s and 2.5 m/s, and the experiments were conducted for the sliding distance of 10 km at room temperature. The coefficient of friction was measured as a function of sliding speed and load. To assess the wear performance under high load capacity, polymer bearings were tested for a wide range of PV factors. The composite ring was weighed before and after the wear tests. Tests were repeated three times and the standard deviation data are presented as error bars. Surface roughness (R_a) and hardness of the steel block were measured as 0.2 ± 0.01 μm and 62 HRC,

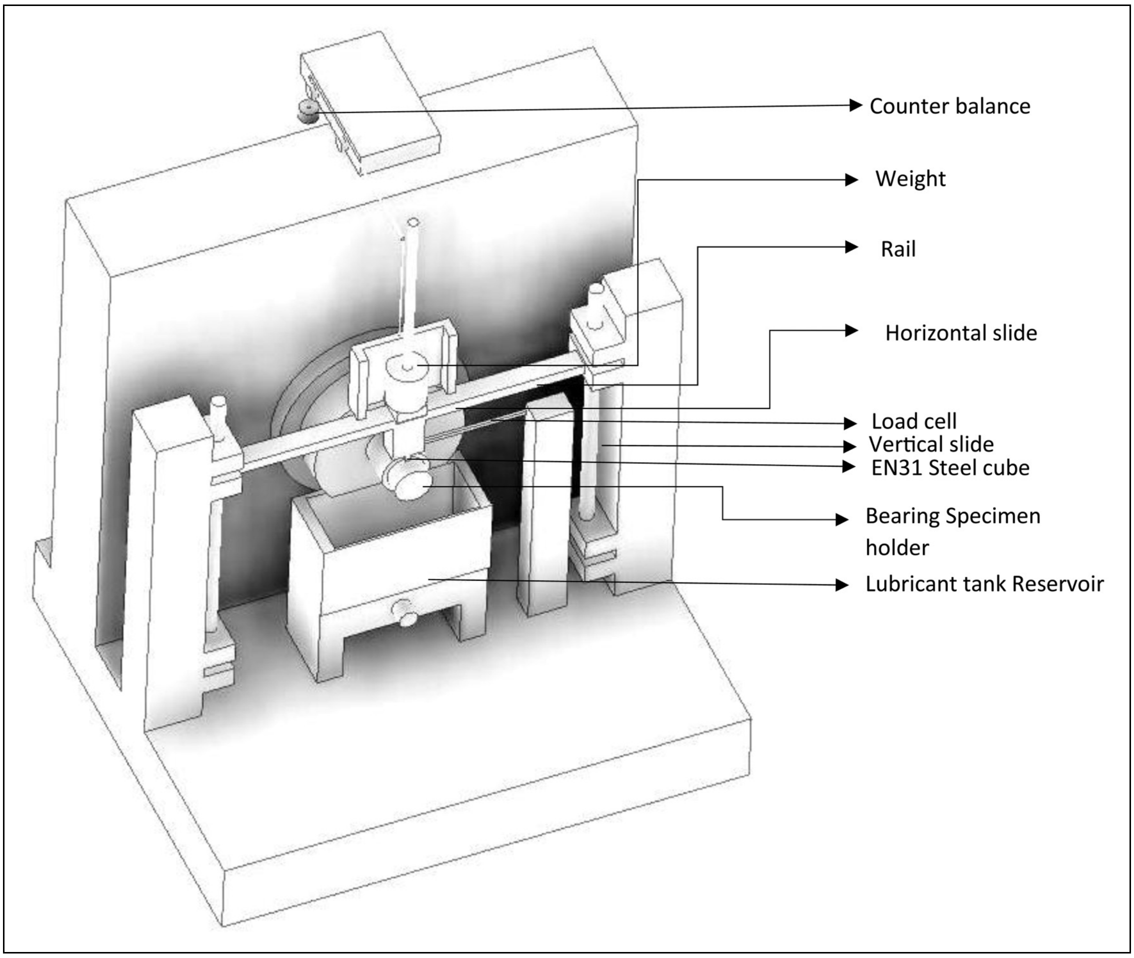

Figure 3. Systematic CAD model of TR 31 lubricity tester.

respectively. Mass loss method was used for measuring the specific wear rate (SWR) of polymer composites. $S.W.R = \frac{\Delta m}{density\ X\ load\ X\ distance}\ mm^3/Nm$. Here S.W.R is the specific wear rate and Δm is the mass loss due to sliding. Interfacial temperature between the tribo-pair during sliding was continuously measured using a non-contact infrared temperature gun.

Mechanical and thermal analysis

Compression tests of polymer composite bearings were performed using a universal testing machine (manufactured by SHIMADZU, Japan) with a cross-head speed of 2 mm/min. Three repeats of the tests with averaging and standard deviation calculation were followed. Micro-hardness of all the composites was evaluated by Vickers hardness tester at room temperature. The data reported are averages of 5 measured values on three different specimens. Glass transition temperature was measured using DMA (Dynamic Mechanical Analysis) instrument (DMA Q-800, TA instruments, USA). Composites prepared for thermal analysis were of dimensions 35 mm × 12.75 mm × 3 mm. In the DMA, composites were tested under single cantilever mode with heating ramp of 5 °C/min and temperature range from 30 °C to 200 °C, at a frequency of 1 Hz and in nitrogen environment. For thermal stability study of the polymer composites, TGA (Thermo-gravimetric analysis) was conducted on Perkin Elmer Thermo-gravimetric Analyzer 4000. For this, a small amount of each polymeric sample, about 3–4 mg, was heated from 50 °C to 800 °C at a heating rate of 10 °C/min in the air atmosphere. Mass loss of sample was recorded as a function of temperature. Reduction in the mass is an indication of thermal degradation of the polymer by chemical reaction, mainly oxidation.

Contact angle and surface energy

The contact angle and surface energy of all the three polymer composites were measured by a goniometer (DSA 25E, Krüss GmbH, Germany) and calculated using the Fowke's method. Two liquids, one polar (distilled water) and one non-polar (n-hexane), were used to measure surface energy. Droplets of volume 0.5 µl of distilled water and n-hexane were used for the contact angle measurement and to minimise the effect of drop impact due to weight of the liquid. For each measurement, 5 replicate data were taken on three different samples, and then the average values and standard deviations were calculated. The variation in water contact angles at several locations of a sample was within ±5°. Tests were conducted in air at ambient temperature and relative humidity of 23 °C and 60%,

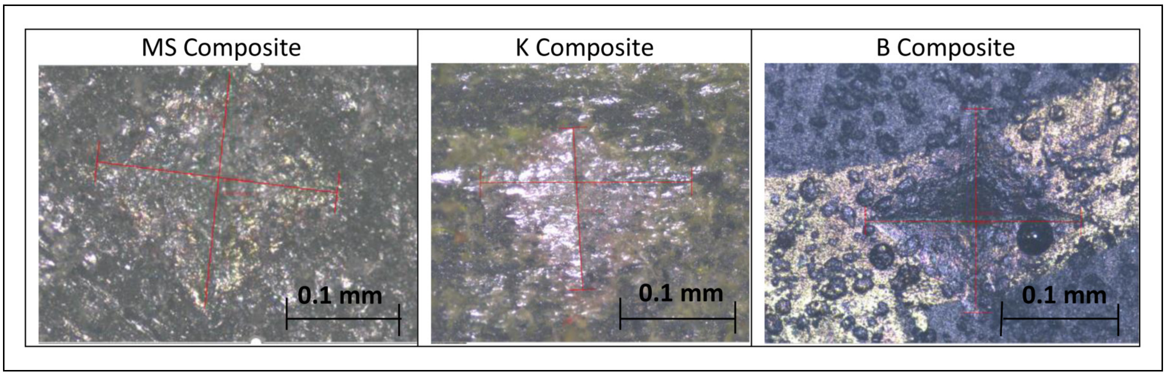

Figure 4. Polymer composites indented by a square pyramidal indenter for micro-hardness test.

respectively. The water contact angle measurement and surface free energy calculation were also conducted on the wear tracks after tribological tests.

Results and discussion

Roughness and density measurement

Polymer composite bearings (MS, K and B) fabricated in this study were polished with fine abrasive paper for uniform line contact. Surface roughness (R_a) of the counter surface EN31 steel block and polymer bearing was measured as $0.2 \pm 0.01\,\mu m$. Roughness of the samples was measured by an optical 3D surface profiler. Archimedes principle was used to determine the density of each sample. The density of composite MS (ternary composite epoxy-UHMWPE-MoS_2) was measured as 1.18 g/cm^3 at 23 °C. The density of composite K (four component composite epoxy-UHMWPE-MoS_2- Kevlar) was measured as 1.13 g/cm^3. The density of composite B (ternary composite epoxy-UHMWPE-MoS_2 with in-situ base oil) was measured as 1.09 g/cm^3. Density of composite 'B' was found to be the lowest among all composites.

Compression and hardness tests

To measure the relative mechanical strengths of the prepared polymer bearings, ring compression test (diametrically loaded) was performed. From displacement vs force graph, average compressive force at failure of each sample was measured. At the failure point, the compressive force of composite 'MS' was found to be 1.7 ± 0.2 kN, compressive force of composite 'B' was 1.6 ± 0.2 kN and the compressive force of composite 'K' was 2.5 ± 0.32 kN (values after $\pm$ indicate standard deviations). From the above data, it was observed that adding Kevlar short fibres in the epoxy-based composite enhanced the compressive strength of the prepared bearing by approximately 47%. This is because Kevlar has a unique combination of high strength, toughness, high modulus and good bonding between the fibre and epoxy matrix. Figure 4 provides images of the indents on the polymer composites after the hardness test at room temperature. Colour wise they are of the same black colour because of the effect of MoS_2 powder which is black in colour. Hardness value of MS, K and B samples at room temperature is shown in Figure 5. Comparing the hardness values, it shows that at room temperature, hardness of composite K is the highest among others. This is another confirmation of the increase in the bulk mechanical strength of the composite after adding short fibres of Kevlar (2 vol%). Adding liquid lubricant (base oil) decreases the hardness of composite which is also in agreement with the decrease in the compressive fracture strength.

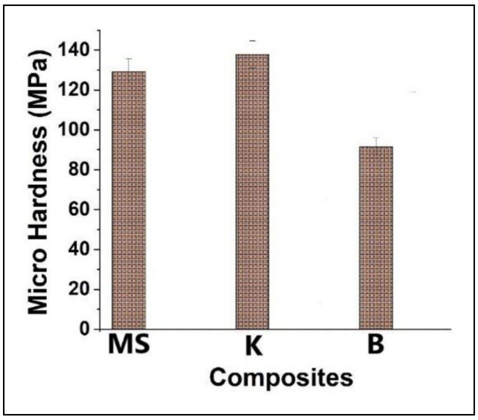

Figure 5. micro-hardness of polymer composites at normal indentation of 10.20 N at room temperature.

Thermal analysis

Figure 6(a) presents the peak of tan δ curve obtained from the DMA test, which signifies the glass transition temperature (T_g). The temperature at which the polymer turns from hard Hookean solid to rubbery state as the temperature is increased, is termed as the glass transition temperature. The T_g of the polymer composite MS is 110 °C. When in-situ base oil is mixed with this polymer composite, T_g decreases from 110 °C to 94 °C as seen for composite B. The reason behind this is the nature of the liquid lubricant that accumulates the heat generated within itself which hinders the thermal conductivity of the polymer composites and softens the polymer at a

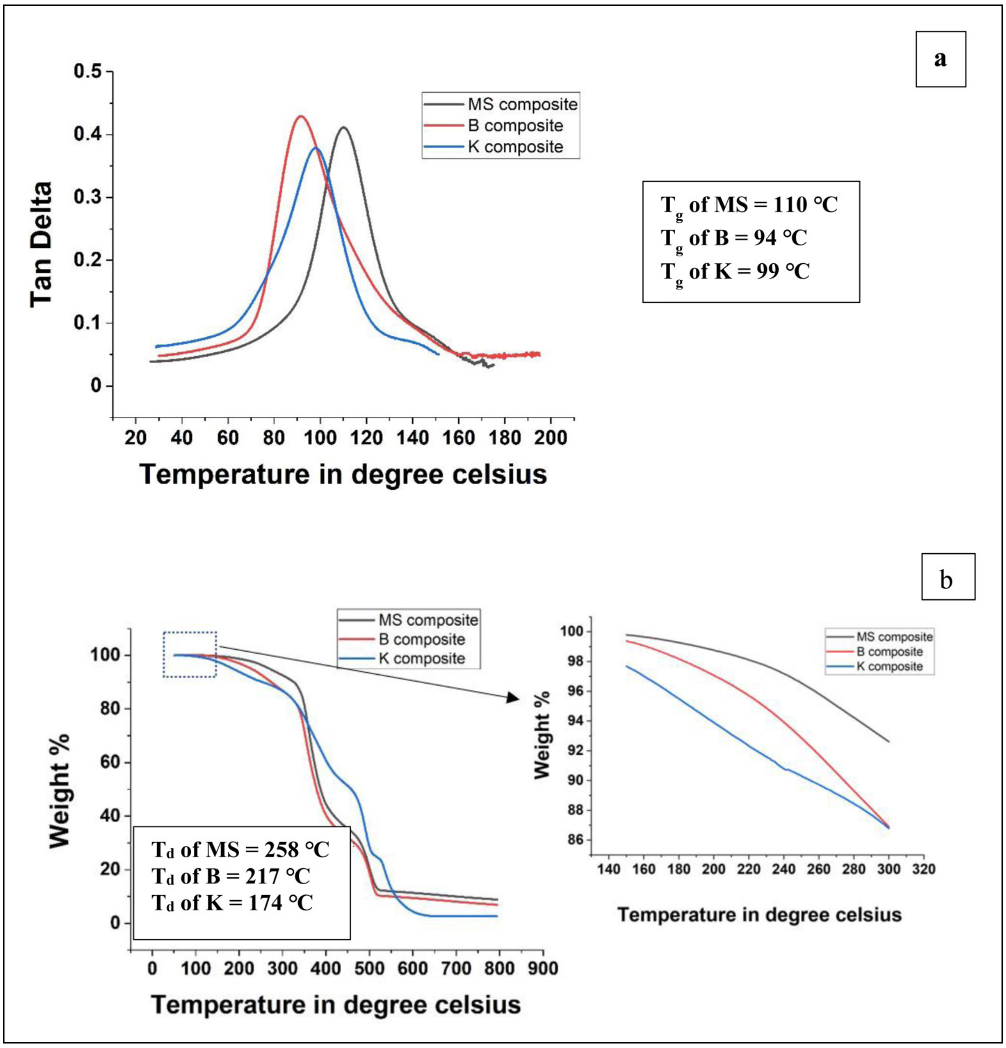

Figure 6. (a) Thermal analysis of polymer composites measured through DMA under single cantilever mode with heating ramp of 5 °C/min and temperature range of 30 °C to 200 °C, frequency 1 Hz in nitrogen environment **(b)** TGA data of polymer composites heated from 50 °C to 800 °C at the heating rate of 10 °C/min in air.

lower temperature. T_g of polymer composite K is 99 °C which shows that Kevlar provides rigidity and strength at normal temperature, however, at higher temperature there is decrease in the bulk mechanical properties because of softening. Lower T_g has the benefit of higher ductility at the normal temperature. Figure 6(b) compares the thermal stability or degradation temperatures of MS, B and K polymer composite bearing in air. Here, T_d (thermal degradation temperature) and thermal stability corresponds to the temperature up to which there is no or negligible mass loss of the polymer composites. T_d of MS, B and K composites are 258 °C, 217 °C and 174 °C, respectively. The lowering in the thermal stability of oil-filled 'B' (217 °C) in comparison to 'MS' (258 °C) is because of the in-situ liquid lubricant base oil which accelerated the oxidation and degradation due to the presence of hydrocarbon oil. It is interesting that Kevlar decreased the T_d for the composite even though Kevlar fibres in general have high thermal degradation temperature in the range of 400–450 °C.[23] This requires further investigation.

Friction and wear results at room temperature

The coefficient of friction (μ) between two interacting members majorly depends upon relative speed, temperature, contact pressure, state of lubrication, the roughness and the nature of the contacting surfaces such as the surface energy and hardness. From the machine design and performance points of view, load and speed are the two dominant factors that affect the tribological properties of materials when other factors are fixed by design. Figure 7(a), (b) compares the coefficient of friction (COF) and SWR of 'MS' polymer composite at varying speeds (0.5 m/s to 2.5 m/s) and loads (20 N to 70 N). Data indicate that as speed increases with load, SWR increases except at low speed (0.5 m/s) and low load (20 N) conditions. At higher loads and speeds, friction tends to decrease. This is quite consistent with previous work which shows that for polymer composites, higher sliding speed would reduce the coefficient of friction, but wear would increase due to interfacial heating and material softening.[24] With increasing load at a given

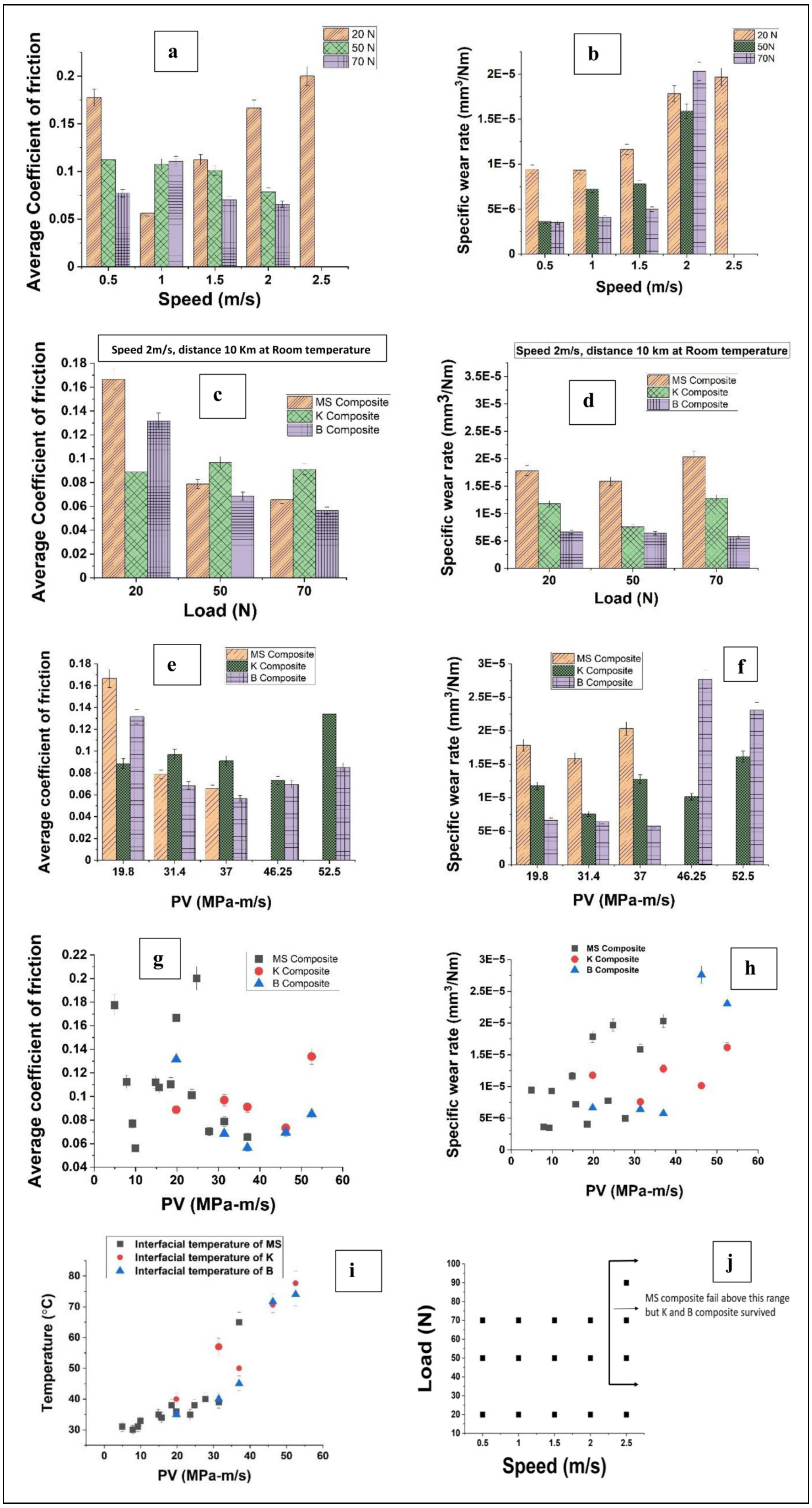

Figure 7. **(a)** Coefficient of friction and **(b)** Specific wear rate of 'MS' composite at different load (20 N, 50 N, 70 N) and variable speed (0.5 m/s – 2.5 m/s). **(c)** Coefficient of friction and **(d)** Specific wear rate of different polymer composite ('MS', 'K', 'B') at different load (20 N, 50 N, 70 N), speed 2 m/s and sliding distance 10 km. **(e)** Coefficient of friction and **(f)** Specific wear rate of different composites at variable PV (19.8 MPa-m/s to 52.5 MPa-m/s) for sliding distance 10 km. **(g)** Coefficient of friction and **(h)** Specific wear rate of polymer composites at wide range of PV (9.9 MPa-m/s - 52.5 MPa-m/s). **(i)** Change in interfacial temperature of different polymer composites with respect to PV (9.9 MPa-m/s - 52.5 MPa-m/s) **(j)** Failure map of composites. All the above data are at room temperature.

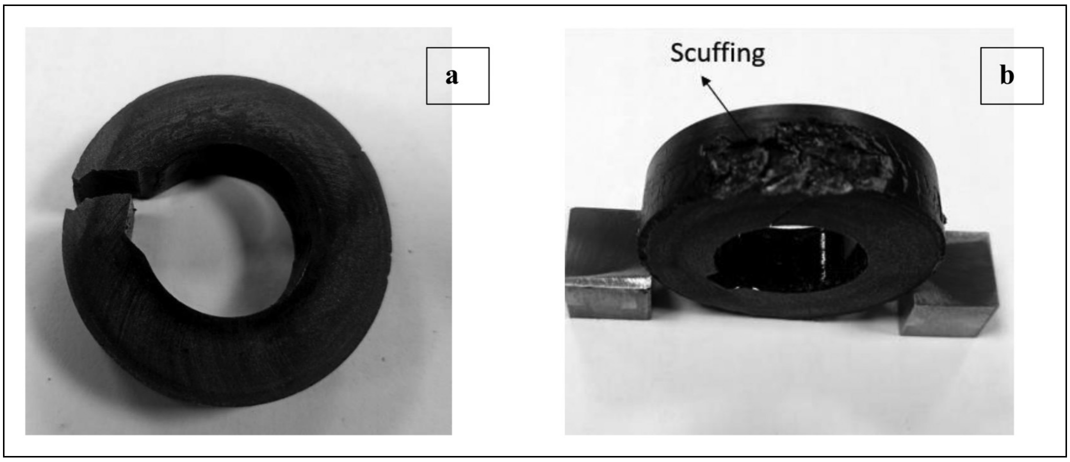

Figure 8. (a) Failed 'MS' composite at 50 N load and 2.5 m/s (b) Scuffing in 'MS' composite due to frictional heating and high adhesive wear at 50 N load.

speed, the coefficient of friction and SWR decreased at lower speeds. However, as the speed is increased the SWR will eventually increase owing to interfacial heating and softening. At high speed of 2.5 m/s and high loads (50 N and 70 N), friction and SWR of 'MS' composite could not be evaluated since the composite failed at this point due to high interfacial temperature and high roughness as seen in Figure (g, h, i,) and Figure 8(a, (b)). In general, as the load increases the coefficients of friction tends to decrease. The lowest coefficient of friction and SWR obtained for 'MS' composite are 0.056 and 9.3×10^{-6} mm^3/Nm, respectively, at 1 m/s and 20 N. Formation of transfer layers on the counterface and the reorientation of MoS_2 basal layers parallel to the sliding surface provide a low-friction interface. The presence of oxidizing agents, particularly water vapour and oxygen, tends to reduce the wear life of the MoS_2 film.[25] Since this composite failed at high speed (2.5 m/s) and high load (50 N and 70 N), polymer composites 'K' and 'B' were tested subsequently.

Figure 7(c) and (d) compares the coefficient of friction and SWR, respectively, of three polymer composites 'MS', 'K' and 'B' at 20 N, 50 N and 70 N normal loads at a fixed speed of 2 m/s for 10 km sliding distance at room temperature. Figure 7(c) shows a decreasing trend for the COF as load is increased. Polymer composite 'B' demonstrates the lowest COF at the highest load and higher range of speed, when compared with 'MS' and 'K'. Figure 7(d) shows that for composite 'B' the SWR shows decreasing trend with normal load. For composites 'MS' and 'K', there is a minimum value at the normal load of 60 N and the SWR increases beyond this load. Overall, the composite 'B' provides the best COF and SWR values when compared with the other two composites within the used range of load and sliding speed. The lubricating effect of liquid base oil at the interface is very essential for excellent tribological performance of polymer composites.

The transfer of the composite material to the counterface initiates substantial wear which is a result of adhesion between the sliding materials and the transfer of the composite material to the steel counterface. In Figure 8(b), the worn surfaces on the MS polymer composite show parallel scratches indicative of scuffing. With low thermal conductivity, polymers cannot dissipate the heat generated at the interface efficiently and hence the surfaces either locally melt or degrade leading to failure. This is the reason that polymers are generally used under low PV factor. PV $_{limit}$ value of a bearing material indicates the utility of the polymer composite under a given set of operating conditions of pressure and speed. Figure 7(e) presents the coefficient of friction as a function of PV value for three different polymer composites 'MS', 'K' and 'B' at five different values of PV (ranging from 19.8 MPa-m/s to 52.5 MPa-m/s). At low PV, COF of MS is high (0.166) and as the PV value increases, COF of MS decreases to 0.06. Beyond the PV of 37 MPa-m/s, 'MS' composite failed due to overheating (65 °C) at the interface of polymer composite and counterface as shown in Figure 8(a) and (b). For composite 'K', in the PV range of 19.8 MPa-m/s to 46.25 MPa-m/s the coefficient of friction is almost constant from 0.09 to 0.07. At

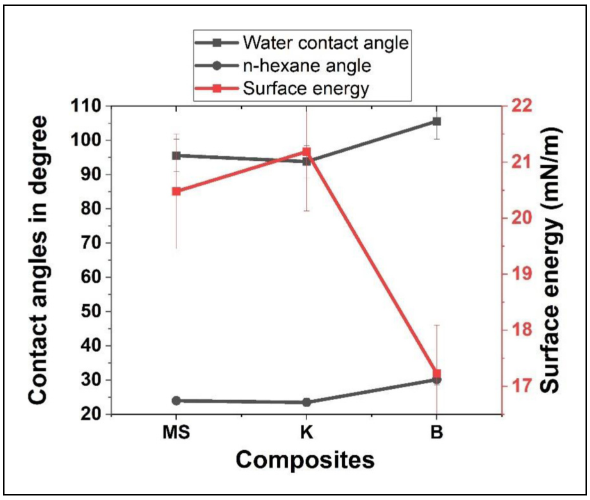

Figure 9. Contact angle and Surface energy of polymer composites after tribology test.

52.5 MPa-m/s the friction was high at 0.13. For composite ‘B’, with increase in the PV value the coefficient of friction decreases till 37 MPa-m/s and increases afterwards till 52.5 MPa-m/s. Initially for PV 19.8 MPa-m/s, COF was 0.13 and at 37 MPa-m/s the lowest COF of 0.056 was observed. It was very interesting to note that ternary polymer composite epoxy-UHMWPE-MoS_2 (MS) failed after 37 MPa-m/s i.e., PV $_{limit}$ for composite ‘MS’ is 37 MPa-m/s but, the four-component polymer composites epoxy-UHMWPE-MoS_2- Kevlar (‘K’) and epoxy-UHMWPE-MoS_2- base oil (‘B’) did not fail till the high PV value of 52.5 MPa-m/s, which was the limit of the present tester. Figure 7(f) presents their corresponding SWR data. The SWR of ‘B’ polymer composite is the lowest (= 5.77×10^{-6} mm^3/Nm) among others at 37 MPa-m/s. For all the composites, as PV value increases SWR also increases. ‘MS’ and ‘K’ composites have high wear rates with respect to ‘B’ composite up to 37 MPa-m/s, and afterwards, wear of ‘B’ is more in comparison to ‘K’ whereas ‘MS’ fails beyond 37 MPa-m/s. Hence, comparing the three composites, ‘K’ performed the best in terms of the SWR at high PV factors and composite ‘B’ performed the best at low PV factors. Figure 7(g) shows the coefficient of friction for a wide range of PV values (9.9 MPa-m/s to 52.5 MPa-m/s) for all the three polymer composite bearings, and Figure 7(h) shows their corresponding specific wear rates. Figure 7(i) presents the interfacial temperatures measured on the surface of the composite at different PV values. Variation in temperature was observed during the tribological experiments. It was found that initially the interfacial temperature between the

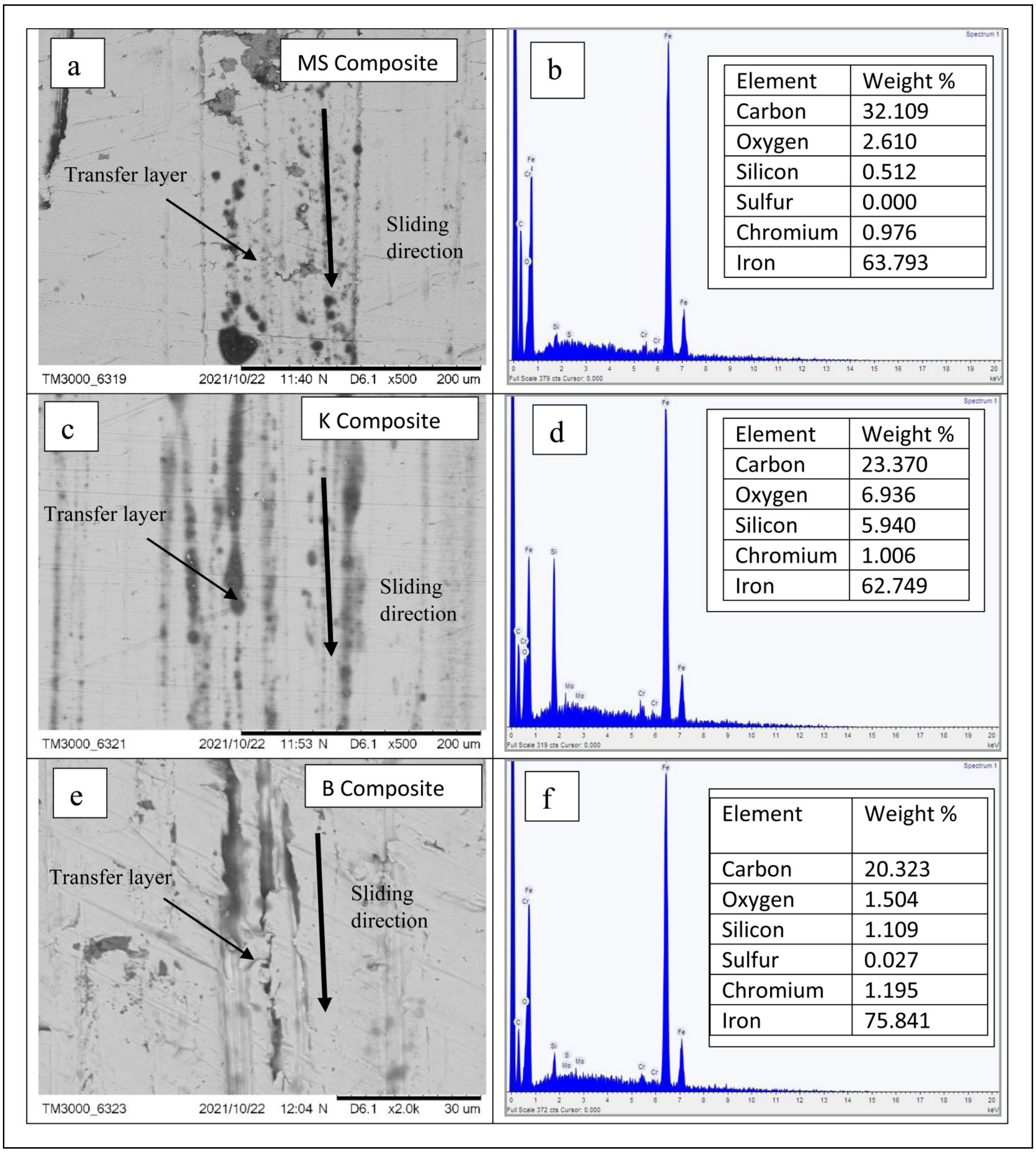

Figure 10. **(a)**, **(b)** shows the SEM image and the corresponding EDX of steel block on which transfer layer of ‘MS’ composite was present on the wear track. **(c)**, **(d)** SEM image and the corresponding EDX of steel block on which transfer layer of ‘K’ composite was present on the wear track. **(e)**, **(f)** SEM image and the corresponding EDX of steel block on which transfer layer of ‘B’ composite was present on the wear track. **(g)**, **(h)** SEM and EDX of rubbed ‘MS’ composite. **(i)**, **(j)** SEM and EDX of rubbed ‘K’ composite. **(k)**, **(l)** SEM and EDX of rubbed ‘B’ composite. All the SEM and EDX of sample are after tribology tests. (continued)

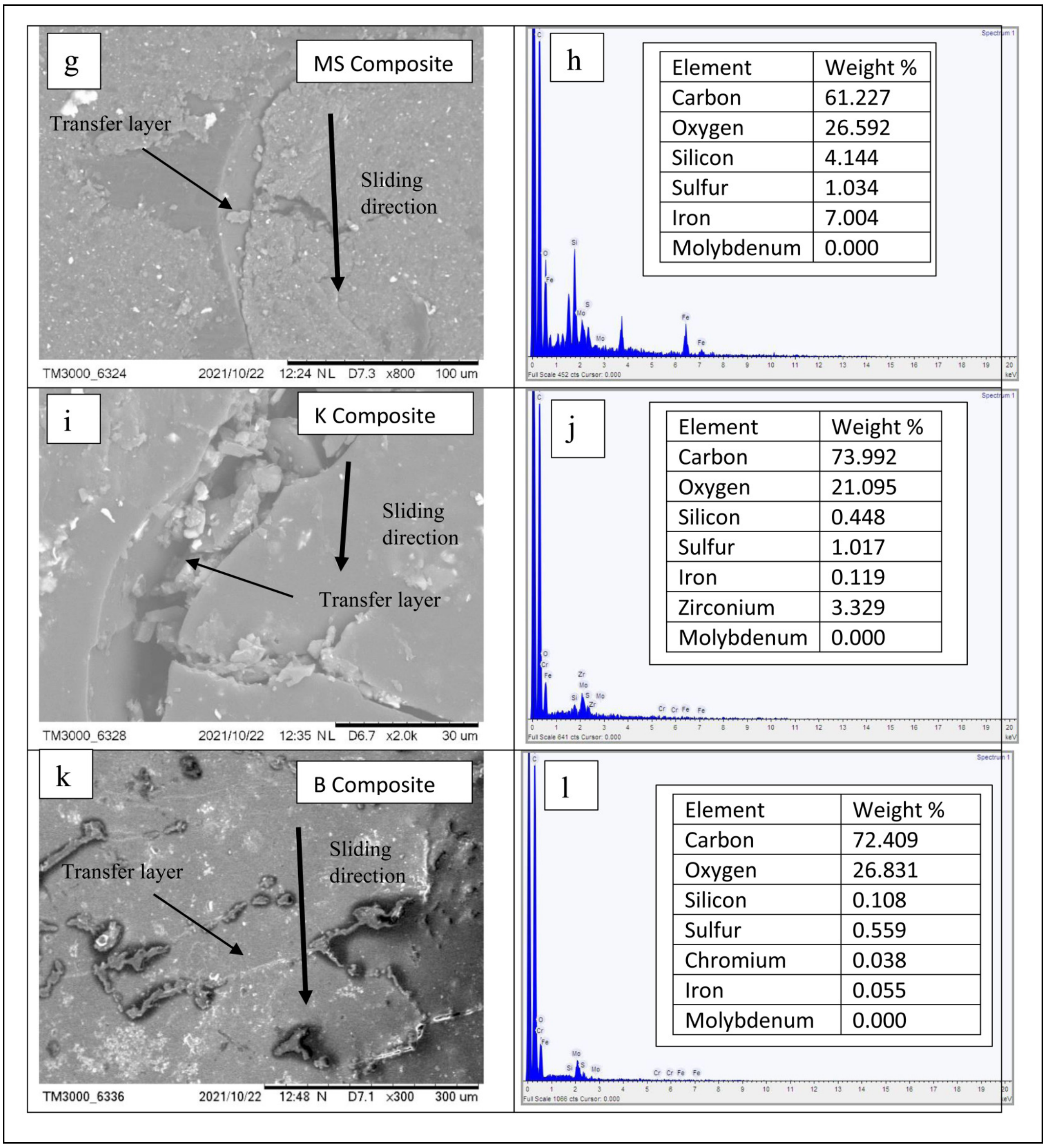

Figure 10. Continued.

composite and the steel block increased monotonously as the PV value was increased. However, at PV value of approximately 37 MPa-m/s, there was sudden increase in the interfacial temperature and beyond this PV, the interfacial temperature again increased in linear fashion but at a greater rate than before. It shows that at 37 MPa-m/s PV, there is drastic change in the surface interaction leading to sudden temperature rise. The composite 'MS' failed at this PV value due to high adhesive wear and scuffing which is, in part, a result of high interfacial temperature. At high PV values, the interfacial temperatures for the composite 'K' and 'B' and the counterface were above 70 °C due to frictional heating which shows that 'K' and 'B' can bear high interface temperature in actual applications. Frictional heating increases the temperature of the component especially in situations where there is limited possibility for heat removal by conduction or convection. As the temperature of polymer increases, the material becomes softer and thus the wear rate also increases by the adhesive wear mechanism.

Overall, it can be concluded that composite B has performed well in terms of friction due to the presence of in-situ base oil and composite K has performed well in terms of SWR due to better bulk mechanical properties.

Figure (j) maps the test conditions of the three composites with respect to load, speed and the failure conditions. It is important to know that composites 'K' and 'B' did not fail at the tested conditions, however composite 'MS' failed at the speed of 2.5 m/s and the load of 50 N and above. The given maximum load and speed were the limitations of the present tester and the composites 'K' and 'B' could be tested for even higher PV value in the future. Figure 8(a) and (b) are the images of the failed 'MS' composite bearings. The images show that the failure happened because of softening, adhesion and removal of the material in large chunk by the mechanism of scuffing. This wear failure could be avoided for composite 'B' by incorporating base oil and reducing friction, and for composite 'K' by incorporating Kevlar fibre and increasing the strength.

Contact angle and surface energy analysis

Figure 9 presents the water and n-hexane contact angles and surface free energies of the surfaces of different composites. Contact angle of the composite 'B' has higher contact angle and lower surface energy when compared to composites 'MS' and 'K'. After adding 17 vol % in-situ base oil in ternary polymer composite epoxy-UHMWPE-MoS_2, the nature of surface changes from hydrophilic to hydrophobic. Due to hydrophobic nature of the surface, the wettability of polymer composite with distilled water and n-hexane is low. It shows low adhesion properties of the polymer. The surface energy of composite 'B' is the lowest (17.23 mN/m). Thus, low surface energy leads to low adhesion and low friction between the surfaces which was observed in the tribological tests (Figure 7 (c) and (e)).

SEM and EDX analysis of the pin surface and the wear track

After the tribology tests, the wear tracks on the EN 31 steel block and on the polymer composites consisted of multiple parallel scratches or grooves on visual observations. The optical microscope images of the wear tracks on the EN 31 steel block and on the composite samples are shown in Figure 10. Figure 10 (a) and (b) show the SEM image and the corresponding EDX of the steel block on which transfer layer of 'MS' composite was deposited on the wear track. Figure 10 (c) and (d) are the SEM image and the corresponding EDX, respectively, of the steel block on which the transfer layer of 'K' composite was present on the formed wear track. Figure 10 (e) and (f) present the SEM image and the corresponding EDX, respectively, of the steel block on which the transfer layer of 'B' composite was deposited on the wear track. The wear tracks on the steel block after friction test shows different compositions containing carbon, sulphur, silicon, oxygen, chromium, iron as seen in EDX analysis (Figure 10). It is seen that the carbon percentage on the wear track is very high with small percentage of molybdenum. The wear tracks on the polymer composite surfaces with scratched parts are shown in the SEM images (Figure 10 (g), (i), (k)) and their corresponding EDX in Figure 10 (h), (j), (l). The C percentage on the composite surface is much higher with very small peak for Mo because of small percentage of Mo in relative terms. After sliding under a PV value of 37 MPa-m/s, wear debris are produced due to abrasion as observed in the SEM images. Wear was started by the initial removal of polymer material from the pin onto the EN31 steel block. The transfer film of polymer composite component materials first forms on the metal pin surface then interacts with the polymer composite bearing. This leads to adhesive interaction (between the transfer film and the composite surface) which produces wear debris. Transfer film on the composite surface often shows some cracks which indicates that the film is not continuous. SEM and EDX analysis on the wear surface of 'MS', 'K' and 'B' polymer composite bearings demonstrate the transfer of iron particles from the steel disc. The relative % of iron concentration on the surface of 'MS' was highest and of 'B' was the lowest. This indicates that higher friction of the polymer composite against the metal surface also has the tendency for higher metal transfer to the polymer composite surface.

Conclusions

Tribological, Mechanical and thermal performances of epoxy-based composites filled with UHMWPE, MoS_2 and base oil or Kevlar were systematically studied under different sliding conditions. Tests were conducted on a block-on-ring geometric configuration with ring made of the composite materials fabricated in this study. Ring fabricated represented a plain bearing. The load was varied from 20 N to 70 N with variation in contact sliding speed from 0.5 m/s to 2.5 m/s which resulted in a wide range of PV value from 4.9 MPa-m/s to 52.5 MPa-m/s. Performances were studied at room temperature. From the experimental results and the surface analyses, the following conclusions are drawn:

(i) Adding 2 vol% Kevlar to ternary polymer composite epoxy-UHMWPE-MoS_2 enhances the compressive strength by 47% from 1.7 KN to 2.5 KN. Kevlar also increases the hardness of the composite from 111.7 MPa for pure epoxy to 137.8 MPa for the Kevlar-filled composite.

(ii) Thermal degradation property of polymer composite epoxy-UHMWPE-MoS_2 is better than that of other composites. T_d of epoxy-UHMWPE-MoS_2 is 258 °C. T_g of this composite is also higher at 110 °C. Base oil and Kevlar added to epoxy-UHMWPE-MoS_2 decrease both T_g and T_d of the base ternary composite.

(iii) After adding 17 vol % in-situ base oil in the polymer composite epoxy-UHMWPE-MoS_2, the nature of surface changes from hydrophilic to hydrophobic. Epoxy-UHMWPE-MoS_2 with 17 vol % in-situ base oil (composite 'B') is the best composite for tribological applications among 'MS' and 'K' composites for the moderate PV factor. Composite 'B' provided the lowest coefficient of friction ($\mu = 0.056$) and the lowest specific wear rate (5.77×10^{-6} mm^3/Nm) at 37 MPa-m/s when compared with other composites. The optimum PV for composite 'B' for the best performance is 37 MPa-m/s. Composite 'MS' failed at the PV value of 37 MPa-m/s whereas, others (B and K) survived despite having high interfacial temperature.

(iv) Composite K was found to have the lowest wear rate at the higher values of PV factors. Both composites B and K survived the PV factor up to 52.5 MPa-m/s.

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